

Problem 4.18

(a) without tidal force, the condition for not being disrupted is

$$\rho \Delta m \left(\omega^2 - \frac{Gm}{r^3} \right) \leq 0$$

i.e.

$$\frac{\omega^2}{\frac{4}{3}\pi} - \frac{Gm}{\frac{4}{3}\pi r^3} = \frac{\omega^2}{\frac{4}{3}\pi} - G\rho \leq 0$$

$$\rho \geq \frac{\omega^2}{\frac{4}{3}\pi G} = \frac{3\pi f^2}{G}$$

$$= \frac{3\pi \times 30^2}{6.672 \times 10^{-11}} = 1.3 \times 10^{14} \text{ kg/m}^3$$

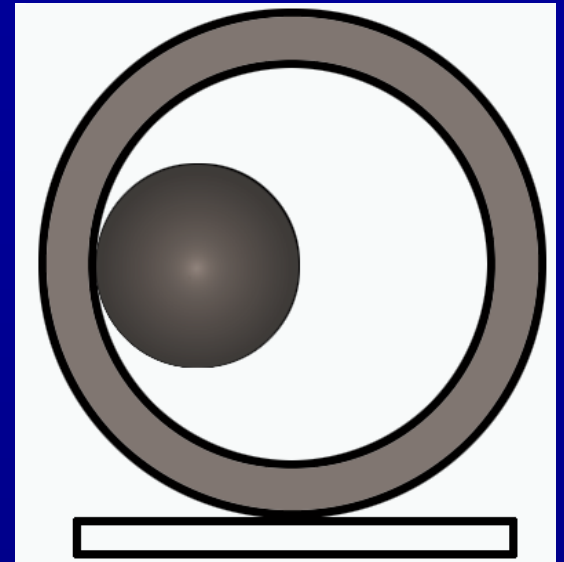
(b)

$$R \leq \sqrt[3]{\frac{m_{\odot}}{\frac{4}{3}\pi\rho}} = \sqrt[3]{\frac{2.00 \times 10^{30}}{\frac{4}{3}\pi \times 1.3 \times 10^{14}}} \approx 1.5 \times 10^5 \text{ m}$$

Problem 5.2

Gravity only changes y_c ; x_c is located $R/2$ left from O , which won't change.

So the shell will move $R/2$ leftward with small ball at the bottom.



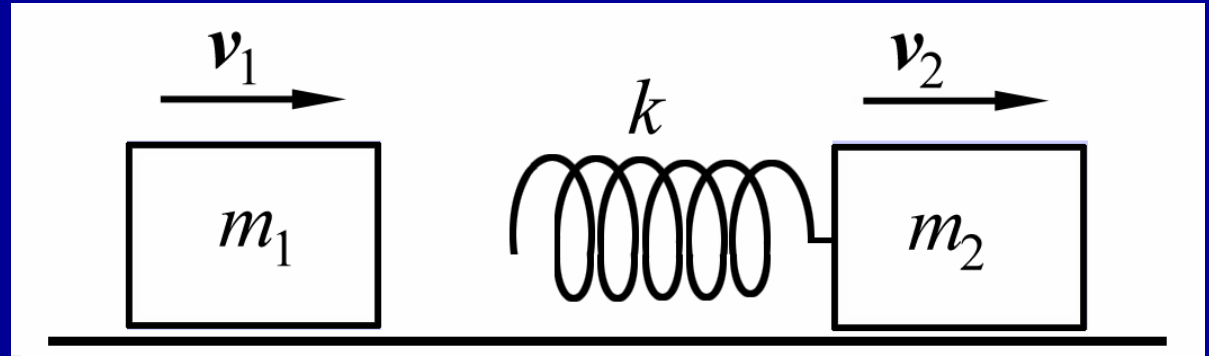
***Problem 5.3** Chain on the table as **system**

$$m \frac{dv}{dt} = mg + (v - 0) \frac{dm}{dt} - N$$

$$m = \rho x \quad v = \sqrt{2gx}$$

$$N = \rho gx + \rho v^2 = 3\rho gx$$

Problem 5.6



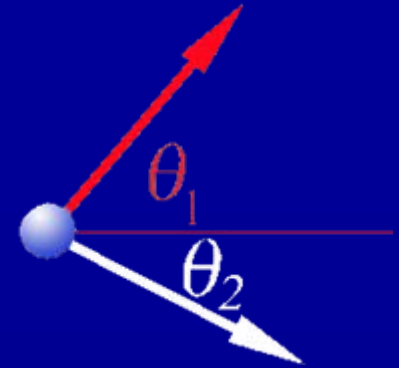
In COM system

$$K = \frac{1}{2}m_1v_{1c}^2 + \frac{1}{2}m_2v_{2c}^2$$

$$K = \frac{1}{2}kx_{\max}^2$$

$$x_{\max} = 0.25m$$

Problem 5.11



Omitting θ_2 and v_2' in conservation equation (p and E), we have

$$\cos \theta = \frac{1}{2} \left[\frac{m_1 - m_2}{m_1} \left(\frac{v_1}{v_1'} \right) + \frac{m_1 + m_2}{m_1} \left(\frac{v_1'}{v_1} \right) \right]$$

As a function of v_1/v_1' , the derivative of $\cos \theta$ leads to

$$\frac{d \cos \theta}{d \left(\frac{v_1}{v_1'} \right)} = \frac{1}{2} \left[\frac{m_1 - m_2}{m_1} - \frac{m_1 + m_2}{m_1} \left(\frac{v_1'}{v_1} \right)^2 \right]$$

Minimum is only possible for $m_1 > m_2$, and

$$\left(\frac{v_1}{v_1'} \right)^2 = \frac{m_1 + m_2}{m_1 - m_2}$$

While

$$\frac{d^2 \cos}{d\left(\frac{v_1}{v'_1}\right)^2} = \frac{m_1 + m_2}{m_1} \left(\frac{v'_1}{v_1}\right)^3 > 0$$

indicating a minimum of $\cos \theta$, or maximum of θ . Substituting the value of v_1/v'_1 into the expression for $\cos \theta$, we have

$$(\cos \theta)_{\min} = \frac{1}{m_1} \sqrt{m_1^2 - m_2^2} > 0$$

or

$$\cos^2 \theta_{\max} = 1 - \frac{m_2^2}{m_1^2}$$

If $m_1 \leq m_2$, $\cos \theta$ is monotonically decreasing function of v_1/v_1' θ can be $(0, \pi]$. Attention should be paid to $\theta = 0$.

For $m_1 = m_2$

$$\mathbf{v}_1 = \mathbf{v}_1' + \mathbf{v}_2'$$

$$v_1^2 = v_1'^2 + v_2'^2$$

So

$$v_1^2 = v_1'^2 + v_2'^2 + 2v_1'v_2' \cos(\theta_1 + \theta_2);$$
$$v_1'v_2' \cos(\theta_1 + \theta_2) = 0$$

None of v_1' , v_2' can be zero except a head-on collision. In that case $v_1' = 0$, $v_2' = v_1$

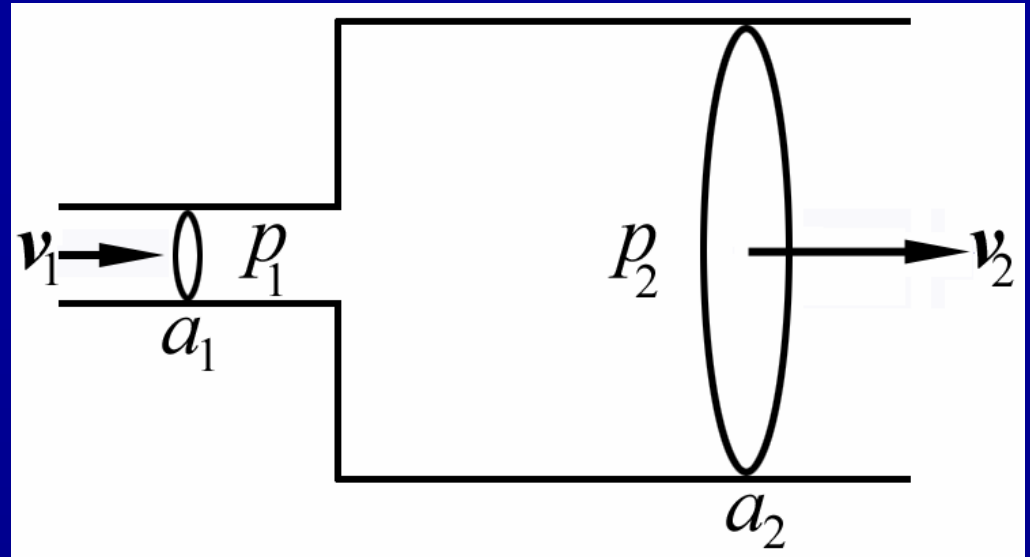
So the only possibility is

$$\cos(\theta_1 + \theta_2) = 0$$

or

$$\theta_1 + \theta_2 = \pi/2$$

*Problem 5.12



$$\begin{aligned}\Delta p &= \rho a_2 v_2 \Delta t \cdot v_2 - \rho a_1 v_1 \Delta t \cdot v_1 \\ &= \rho a_2 v_2 \Delta t \cdot (v_2 - v_1)\end{aligned}$$

$$\begin{aligned}I &= p_1 a_1 \Delta t + \underline{p_1 (a_2 - a_1) \Delta t} - p_2 a_2 \Delta t \\ &= (p_1 - p_2) a_2 \Delta t\end{aligned}$$

From wall

So

$$p_2 - p_1 = \rho v_2 (v_2 - v_1)$$

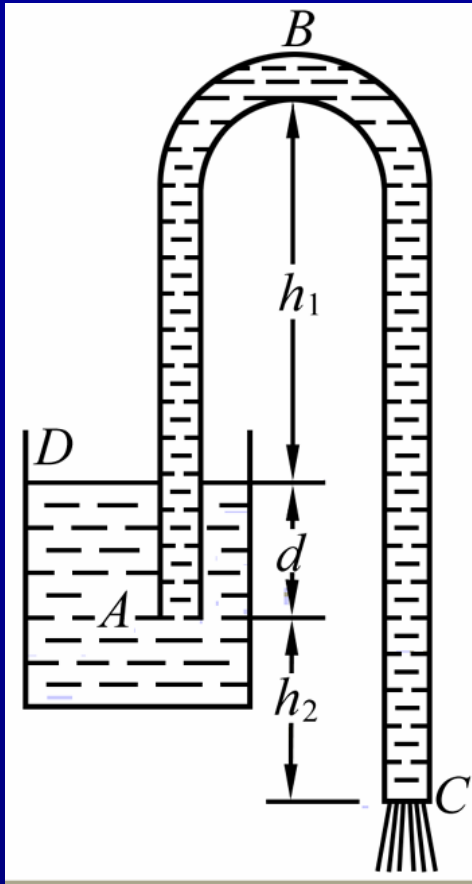
b) For gradually widening pipe, stream lines exist. According to Bernoulli's equation

$$p_2 - p_1 = \frac{1}{2} \rho (v_1^2 - v_2^2)$$

c) Δp lost

$$\begin{aligned} \Delta p &= \frac{1}{2} \rho (v_1^2 - v_2^2) - \rho v_2 (v_1 - v_2) \\ &= \frac{1}{2} \rho (v_1 - v_2)^2 \end{aligned}$$

Problem 5.13



(a) For stream line from D to C

$$p_0 + \rho g(d + h_2) + \frac{1}{2} \rho v_D^2 = p_0 + \frac{1}{2} \rho v_C^2$$

$$v_D = v_C \frac{S_C}{S_D} \ll v_C$$

$$v_C = [2g(d + h_2)]^{1/2}$$

(b) For B to C

$$p_B + \rho g(h_1 + d + h_2) + \frac{1}{2} \rho v_B^2 = p_0 + \frac{1}{2} \rho v_C^2$$

$$p_B = p_0 - \rho g(h_1 + d + h_2)$$

(c) From (b)

$$h_1 = \frac{1}{\rho g} (p_0 - p_B) - d - h_2 \leq \frac{p_0}{\rho g} = 10.3\text{m}$$

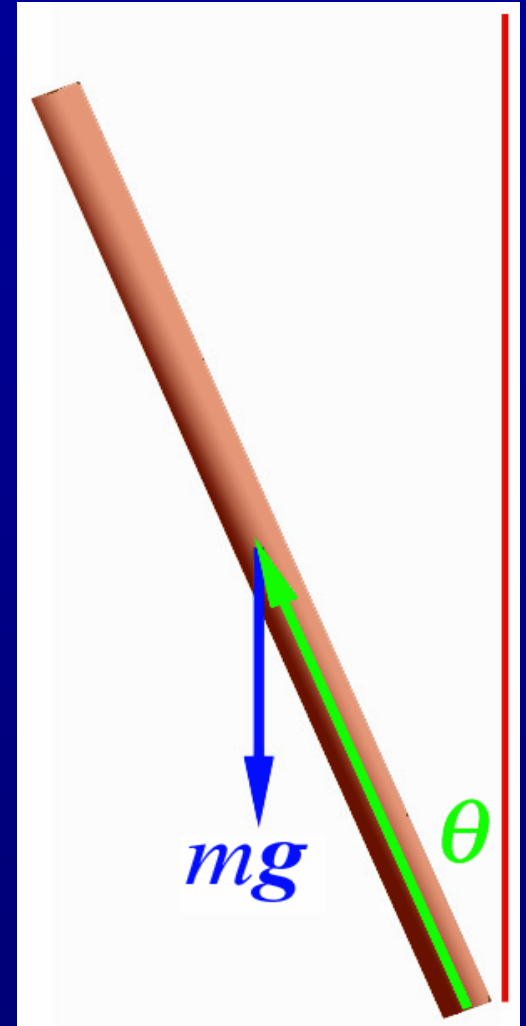
$p_B \geq 0$ **determines the limit on h_1**

Problem 6.2 With respect to O , contact point:

$$\frac{l}{2} mg \sin \theta = \frac{1}{3} ml^2 \frac{d\dot{\theta}}{dt}$$

$$\frac{3}{2} g (1 - \cos \theta) = \frac{1}{2} l \dot{\theta}^2$$

$$l \dot{\theta} \Big|_{\theta=\pi/2} = \sqrt{3gl} = 5.4 \text{ m/s}$$



Or conservation:

$$0 + mg \frac{l}{2} = \frac{1}{2} m \left(\frac{l}{2} \omega \right)^2 + \frac{1}{2} I \omega^2 + 0, \quad mgl = \frac{1}{3} ml^2 \omega^2$$

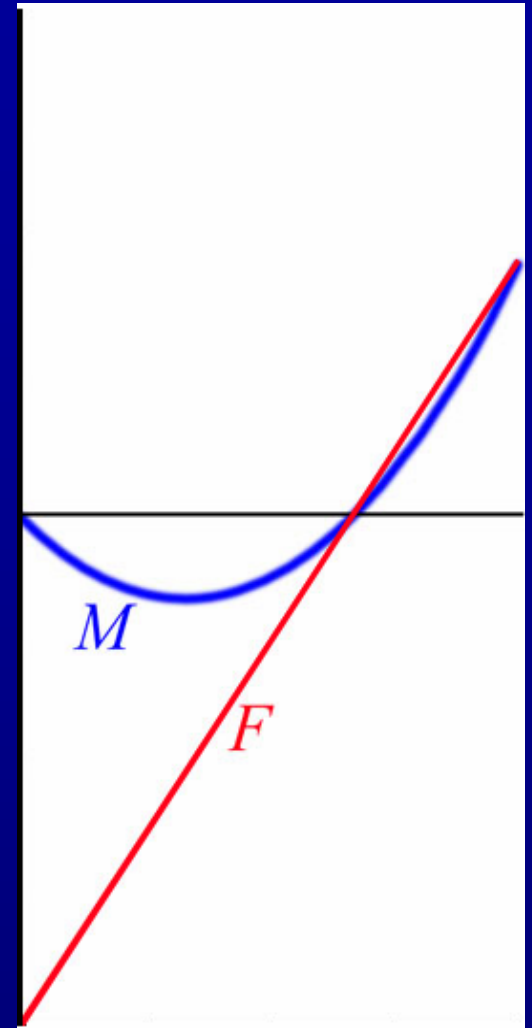
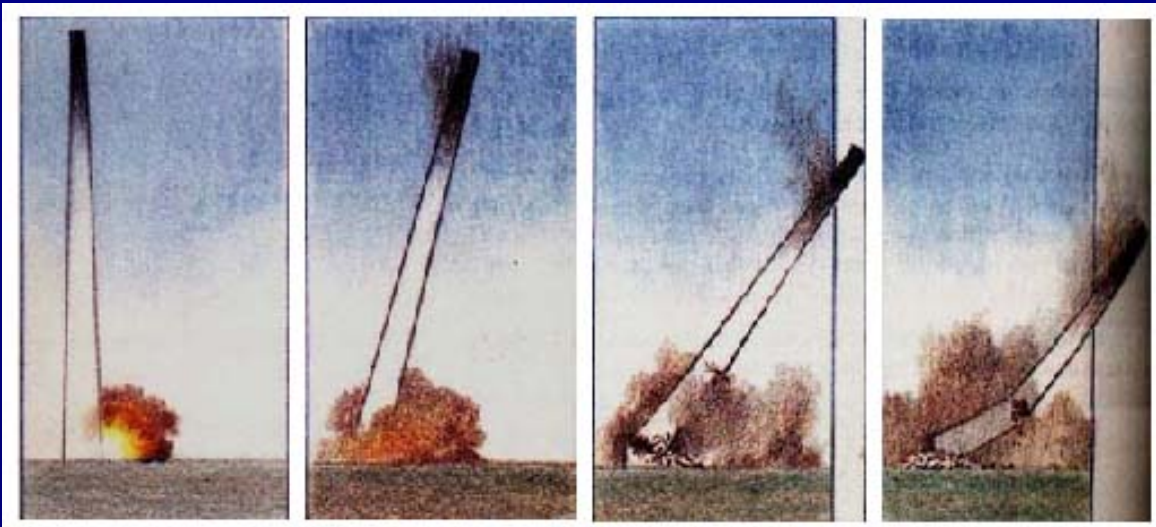
Discussion: in bar system, each segment experiences an inertial force in e_θ direction:

$$-\Delta m \cdot r \ddot{\theta} = -\frac{3}{2} \Delta mg \frac{r}{l} \sin \theta$$

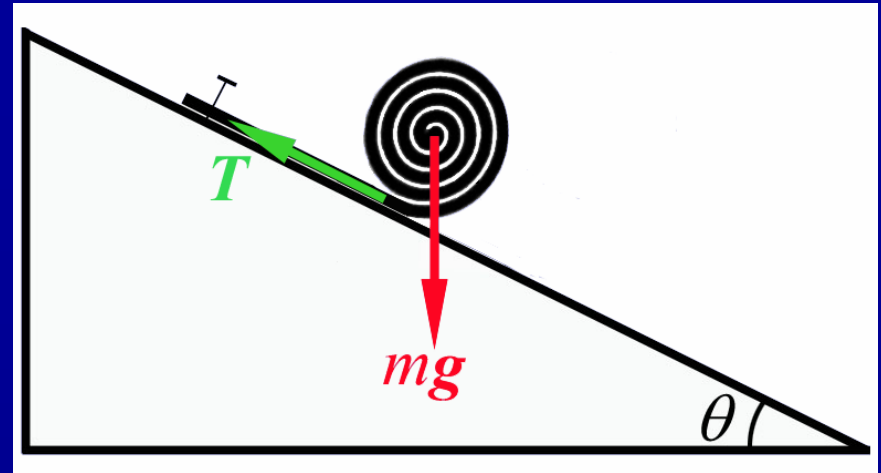
The component of gravity in e_θ direction is $\Delta mg \sin \theta$. The resultant force is then

$$F = -\frac{3}{2l} \Delta m g \sin \theta \left(r - \frac{2}{3} l \right)$$

$$M_{\otimes} = \frac{3r}{2l^2} \Delta m g l \sin \theta \left(r - \frac{2}{3} l \right)$$



Problem 6.3



$$\rho(\ell - x)g \sin \theta - T = \rho(\ell - x)a \quad (1)$$

$$R(x) \times T = \frac{1}{2} \rho(\ell - x)R^2 \times \alpha \quad (2)$$

$$dx = R(x) \cdot d\varphi \quad (3)$$

From Eq.(2) $T = \frac{1}{2} \rho(\ell - x)a$

Substitution of it into Eq.(1) yields

$$a = \frac{2}{3} g \sin \theta$$

The time needed is

$$t = \sqrt{2s/a} = \sqrt{3\ell/g \sin \theta}$$

Discussion: $dx = R(x) \cdot d\varphi$ leads to

$$v = R\omega$$

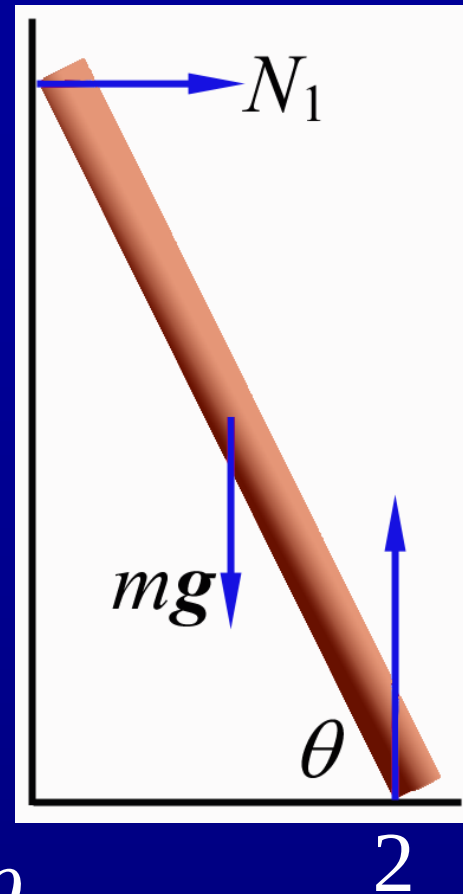
But

$$a = R\alpha + \dot{R}\omega$$

Problem 6.5

Conservation of energy

$$mgL \sin \theta_0 = mgL \sin \theta + \frac{1}{2} mL^2 \dot{\theta}^2 + \frac{1}{2} \left[\frac{1}{12} m (2L)^2 \right] \dot{\theta}^2$$



Motion of com along x : $x = L \cos \theta,$

$$\dot{x} = -L \sin \theta \dot{\theta} \quad \ddot{x} = -L (\cos \theta \dot{\theta}^2 + \sin \theta \ddot{\theta})$$

$$N_1 = -mL (\cos \theta \dot{\theta}^2 + \sin \theta \ddot{\theta})$$

$$N_1 = -mL(\cos \theta \dot{\theta}^2 + \sin \theta \ddot{\theta})$$

Torque equation around 2

$$N_1 \cdot 2L \sin \theta - mgL \cos \theta = \left[\frac{1}{3} m (2L)^2 \right] \ddot{\theta}$$


$N_1 = 0$ leads to

$$\begin{aligned} \dot{\theta}^2 &= -\frac{\sin \theta}{\cos \theta} \ddot{\theta} \\ &= \frac{\sin \theta}{\cos \theta} \frac{mgL \cos \theta}{\frac{4}{3} mL^2} \end{aligned}$$

$$\dot{\theta}^2 = \frac{3g}{4L} \sin \theta$$

Substituting into conservation equation

$$mgL \sin \theta_0 = mgL \sin \theta + \frac{1}{2} mL^2 \dot{\theta}^2 + \frac{1}{2} \left[\frac{1}{12} m(2L)^2 \right] \dot{\theta}^2$$

We have

$$\sin \theta_0 = \frac{3}{2} \sin \theta$$

or

$$2L \sin \theta = \frac{2}{3} \times 2L \sin \theta_0$$